EN.530.646
Robot Devices, Kinematics, Dynamics, and Control
Lab 0 Instructions

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- *This is exclusively used for Fall 2026 EN.530.646 RDKDC students and is not to be posted, shared, or otherwise distributed.*

The labs for RDKDC are built on the Robot Operating System (ROS), which runs on Linux. If you're new to Linux, there are many excellent tutorials available online to help you get started — including this one and this one.

The "Accessing course content via Canvas" section explains how to download lab starter code and updates. Before beginning lab work, you will get familiar with the two types of robots we will be using in this course – UR5 or UR5e – and you need to get familiar with MATLAB. These affect how your environment should be configured. See the "UR5 vs. UR5e" and "MATLAB" sections for guidance.

To set up the lab environment on your personal computer, follow the steps in "Choice 1: Working on PC", which covers installing Ubuntu and ROS2. If you prefer not to install Linux on your own laptop, 12 pre-configured workstations are available in Wyman 170. You can log into these machines with your JHED credentials. See the "Working on Wyman Computers" section for setup instructions.

Once your environment is configured, follow the "Verify Installation" section to ensure everything is working correctly before starting the labs.

Accessing Course Content via Canvas

We use Canvas to distribute course materials. The ROS2 Jazzy setup scripts provided on Canvas will download the RDKDC starter package from Github

All ROS2 code can be found in the following repository:

```
https://github.com/Warrior-hound/rdkdc_jazzy
```

The Lab 0 shell scripts will clone this repository automatically into `~/rdkdc_ws/src/rdkdc`. If the repository is private when you run the script, GitHub may prompt you for access. Use your GitHub ID as your username and a Personal Access Token (PAT) as your password. If the repository is public, no GitHub login is needed.

Homework starter code (if any) will be made visible as each assignment is released. If you have trouble accessing any course repository, please contact the instructors via email.

UR5 vs. UR5e

In Wyman 170, we have two types of collaborative robotic arms: the **UR5** and **UR5e**. While they are similar in appearance and commonly grouped under the "UR5" name, they are in fact distinct models. Both

robots have six joints, giving them six degrees of freedom—that is, six independent ways they can move (such as rotating the base, moving the arm up and down, bending the wrist, etc.). This flexibility allows the robot to position its end effector (e.g., a gripper) precisely in 3D space. You can read more about their technical specs in this link.

In this course, you have the chance to work with both UR5 and UR5e, and we will help you set up your lab environment that works for both models.

Important: Although the robots may look alike, there are subtle differences in their **kinematics** (i.e., the mathematical model of how they move) and their **launch configurations**. As a result, code written for one robot **cannot** be directly used for the other. The course is designed to first teach you how to simulate the robot (UR5 or UR5e), and then how to control the actual hardware in the lab.

Your Coding Language: MATLAB

Lab assignments in this course will use MATLAB. MATLAB is widely used in academia and research—especially for numerical computation, control systems, and robotics due to its powerful built-in toolboxes and ease of matrix manipulation. It is also the language of choice for many robotics textbooks and simulations in educational settings.

How to install MATLAB on Linux can be found here:
https://www.mathworks.com/help/install/ug/install-products-with-internet-connection.html.

When installing MATLAB, make sure to select `ROS Toolbox` and `Robotics System Toolbox`. During installation, check the box to create `symbolic links`. This allows you to launch MATLAB from the terminal by typing `matlab`. Be sure to enter your exact Linux username during installation. Otherwise, you may encounter license issues. If you do run into license-related errors, refer to this helpful Stack Overflow post for troubleshooting.

Working on Personal Computer / Laptop

1 Install Linux

We use Ubuntu 24.04 LTS (Noble) for this course. There are several options for using Linux. One is to install Linux as the main OS of your computer. Another option is to dual-boot your computer. For the dual-boot setup on Windows, see this video. To install Linux, you need to prepare a bootable USB. Here is how to do this on Windows and Mac. Other useful information can be found in this link.

Note that to make a Bootable USB on Windows, you will need to install `Rufus`, whereas on Mac, Etcher is recommended. If necessary, you might need to turn off BitLocker (see here).

Then follow the instructions to install Ubuntu.

NOTE: *Don't forget to look up your computer model.* The process may depend on the machine model. An example, dual boot for a Dell XPS 15 model is found here.

If you are already using an older version of Ubuntu, you can upgrade by following command lines:

```
sudo apt update
sudo apt full-upgrade -y
sudo do-release-upgrade
```

Refer to and compare with the site
https://ubuntu.com/desktop/docs/en/latest/how-to/upgrade-ubuntu-desktop/.
If you opt to upgrade, you may need to uninstall ROS first before upgrading the Ubuntu system. You will see messages on the screen for this. See the following link. You may also get an issue with dpkg returning an error, see here for troubleshooting steps.

Alternatively, you can use a virtual machine such as Parallels, VMware, or VirtualBox. Note that virtual machines can only be used for simulation and cannot be used for controlling the robot later in the course. Recently, there is a personal version of VMware Fusion, which is for Mac OS systems. If using VirtualBox, you may run into an issue where your user account is not given SUDO privilege. If this is the case follow the instructions in this link to fix.

Other possible options include:

- Windows Subsystem for Linux (WSL), which is a good option for Windows users that achieves better performance than virtual machines while not requiring rebooting (like dual boots) to access Linux. Ubuntu comes as the default for this. See instructions here.
- Docker

Note that the two "other possible options" will not be supported by the course instructors. In other words, use it at your own risk.

2 Install ROS2 and Necessary Packages

In order to install ROS2 Jazzy and the necessary packages please follow one of the options below. Option A is recommended, but if you encounter any issues or already have ROS2 Jazzy installed try Option B.

A. Install ROS2 and Necessary Packages with a Single Shell Script

Download the shell script files from the canvas. You will install with '`rdkdc_laptop_with_ros_jazzy.sh`' file in your `Lab0/` folder. Open a new terminal, navigate to the folder that contains this shell script, and run the following lines:

```
$ sudo chmod 755 rdkdc_laptop_with_ros_jazzy.sh
$ sudo ./rdkdc_laptop_with_ros_jazzy.sh
```

B. Install ROS 2 and Necessary Packages Separately

Follow the instructions to install ROS2 Jazzy. Be sure to install the "Desktop" version and not the "ROS-Base" version. Also, be sure to install the "Development Tools".

You will install with '`rdkdc_laptop_jazzy.sh`' file in your `Lab0/` folder. Open a new terminal, navigate to the folder that contains this shell script, and run the following lines.

```
$ sudo chmod 755 rdkdc_laptop_jazzy.sh
$ sudo ./rdkdc_laptop_jazzy.sh
```

It is now successfully installed!

Working on Wyman Computers

This section provides instruction on setting up your work environment if you decide to work in Wyman 170.

1 Install ROS2 Packages

Download '`rdkdc_wyman_jazzy.sh`' file (it is included in the zip file) from the canvas, Lab0 module. Open a new terminal, go to the folder that contains the shell script, and follow the steps to install supporting ROS packages for this course.

```
$ chmod 755 rdkdc_wyman_jazzy.sh
$ ./rdkdc_wyman_jazzy.sh
```

You have successfully installed it!

Verify Installation

Open a new terminal. Type "`ros2 launch rdkdc <ur5/ur5e>_simulation.launch.py`". You should see something like Figure 1.

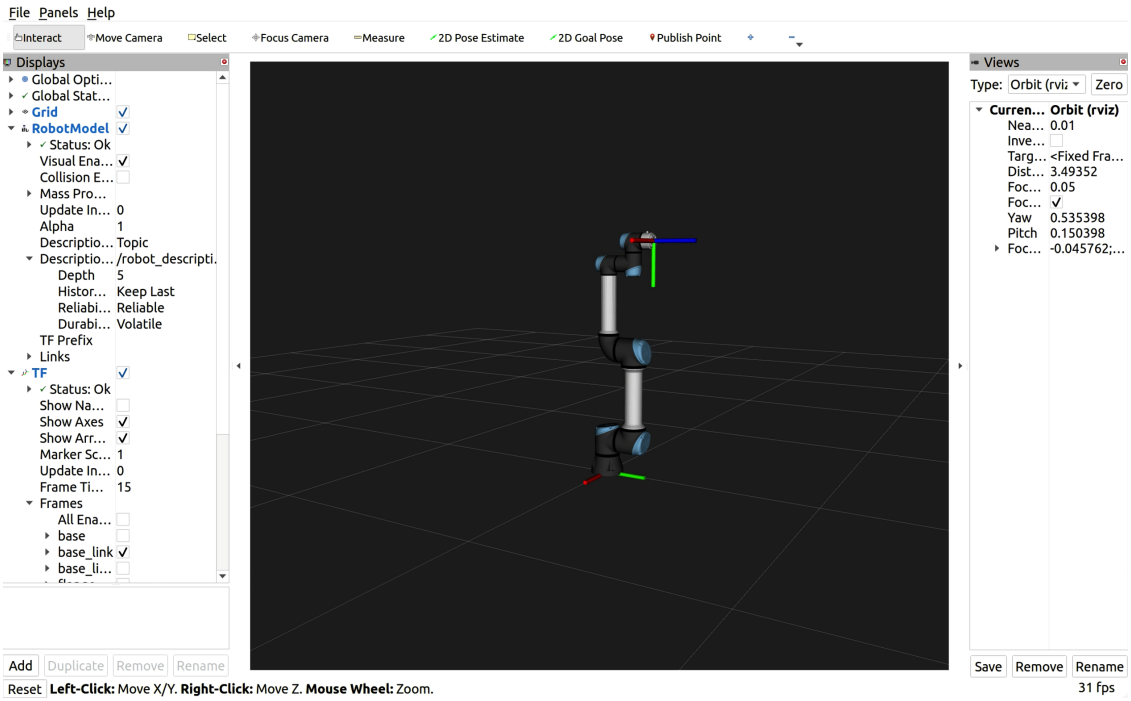


Figure 1: Simulation Window

References

[1] Tobias Fischer, Wolf Vollprecht, Silvio Traversaro, Sean Yen, Carlos Herrero, and Michael Milford. A RoboStack Tutorial: Using the Robot Operating System Alongside the Conda and Jupyter Data Science Ecosystems, *IEEE Robotics and Automation Magazine*, doi:10.1109/MRA.2021.3128367, 2021.